

Automating Year-End Accounting Notes

An AI System to Generate Audit-Ready Financial Statement Notes from
Xero Data

Name: Jatin Arora

Course: B.Sc. (Hons) Artificial Intelligence & Data Science

Student ID: 22830110

Abstract

This dissertation investigates whether an AI-driven system can increase productivity and consistency in year-end accounting note preparation while maintaining professional quality, auditability, and ethical compliance. The project explores an end-to-end workflow that integrates Xero accounting data with an AI-assisted note generation system. The approach combines deterministic numeric reconciliation, provenance tracking, and constrained natural language generation to automate the drafting of key financial statement notes. The research evaluates productivity gains, factual accuracy, and professional acceptability through a human-in-the-loop review with accountants, while addressing anonymisation, ethical boundaries, and general research risk assessment.

1. Introduction

Research question: To what extent can an AI-driven system improve productivity and consistency in the preparation of year-end accounting notes while maintaining professional quality, auditability, and ethical compliance?

Short motivation: Preparing year-end notes is time-consuming and judgement-heavy. An automated system that reliably produces draft notes with clear provenance could materially increase efficiency for accounting firms, reduce routine work for accountants, and improve consistency across clients—while preserving human oversight for judgemental items.

2. Background / Literature Review (summary)

Why the problem is significant:

- Year-end notes (fixed assets, accruals, loans, related parties, contingencies, etc.) are mandatory and require reconciliation of ledger balances, explanations of movements, and disclosure of judgemental events. Producing these notes manually consumes many billable hours.

- Recent advances in large language models (LLMs) and retrieval-augmented generation (RAG) make high-quality text generation from structured data feasible, but challenges remain: factual accuracy, provenance, accounting rules enforcement, and regulatory compliance.
- Studies show AI systems can enhance productivity and output quality when used alongside human professionals rather than as replacements (Brynjolfsson & McAfee, 2017; Kokina & Davenport, 2017). However, audit and accounting automation introduces new ethical and trust-related risks (Issa et al., 2016; Mittelstadt et al., 2016).

Key literature / contextual domains to review:

- Accounting standards and disclosure practice (IFRS / local GAAP guidance on notes).
- Xero API / bookkeeping system data models and constraints.
- Data engineering for financial data (ETL, reconciliation, traceability).
- NLP/LLM methods for data-to-text, retrieval-augmented generation, and hallucination mitigation.
- Explainable AI and human-in-the-loop evaluation in professional decision support systems.

3. Aim and Objectives

Aim: Design, implement and evaluate a prototype system that automatically generates year-end accounting notes from Xero data, providing auditable numeric reconciliations, explanatory narrative, and provenance sufficient for an accountant to review and sign-off.

Objectives:

1. Select and define 3–5 target note types to automate (e.g., Fixed Assets; Trade Receivables; Loans & Borrowings; Revenue Recognition; Prepayments/Accruals).
2. Build a reproducible ETL pipeline to extract, transform and store relevant Xero data.
3. Define accounting mapping tables and deterministic rules to compute the numeric backbone for each note.
4. Develop a provenance model linking every computed figure to the source transactions/journal entries.

5. Implement a note generation module using an LLM (or RAG) with constrained prompts and templates.
6. Design and run a human evaluation with professional accountants to measure factual accuracy, acceptability, and time savings.
7. Analyze results, document limitations, and propose improvements for explainability, ethics, and compliance.

4. Methodology

Approach: Mixed-methods engineering research combining systems engineering (ETL, rules), AI/NLP (LLM / RAG), and human evaluation.

Methods:

- **Data acquisition & pre-processing:** Use Xero sandbox or anonymised firm data. Implement OAuth2 extraction and normalize for deterministic queries.
- **Rules & analytics layer:** Encode accounting rules in Python/dbt. Compute balances, movements, and adjustments with full traceability metadata.
- **LLM integration:** Use a high-quality model (API or hosted) with constrained prompts to ensure numeric and factual consistency.
- **Hallucination mitigation & explainability:** Only permit the model to state numeric facts present in structured input; require provenance for each numeric statement.
- **Human-in-the-loop review:** Accountants review AI-generated drafts, marking edits and feedback to evaluate trust and productivity.
- **Evaluation methodology:** Measure factual accuracy, completeness, edit distance, time savings, and qualitative trust ratings.

5. Implementation

High-level architecture:

1. Data Layer: Xero API → ETL → PostgreSQL / Parquet.
2. Rules Engine: Python/dbt transformations for note computations.
3. Provenance Store: Mapping from computed figures to source transactions.
4. LLM Service: Receives JSON and returns draft notes with provenance.

5. Review UI: Flask/Django web app for accountants to edit and approve.
6. Audit Logging: Records model version, input snapshot, and final output.

Tech stack:

- Python, pandas, SQLAlchemy, Airflow/Prefect.
- PostgreSQL (or parquet data lake).
- Xero Python SDK for API.
- LLM provider (GPT-based or hosted open model).
- Django + React (or server-rendered templates).
- Docker + Git for reproducibility.

6. Testing / Evaluation

Quantitative tests:

- Numeric accuracy ($\geq 99\%$ match with inputs).
- Completeness of required disclosures.
- Edit distance between AI draft and human final.
- Time savings: manual vs. AI-assisted.

Qualitative tests:

- Readability, acceptability, trust (Likert-scale ratings).
- Error categorisation: omissions, numeric mismatches, hallucinations.

User study:

- 6–10 accountants reviewing anonymised datasets and AI drafts.
- Measure perceived usefulness and efficiency gains.

7. Research Risk Assessment

- **Data privacy:** Use anonymised/sandbox data; encrypt local storage.
- **Model hallucination:** Use deterministic numeric rules and provenance-enforced generation.
- **Ethical non-compliance:** Restrict to draft generation; human approval required.
- **Bias or misrepresentation:** Human review and transparency logs.
- **Technical risk:** Use SDKs and fallback models.

A general institutional Research Risk Assessment form (low-risk category) will accompany the dissertation submission.

8. Professional, Social, Economic and Legal Issues

Ethical considerations, anonymisation, and boundaries

All datasets are anonymised or sourced from Xero sandbox. The system’s boundary ends at *draft note generation* — it does not perform audits or final sign-off. Human accountants retain full professional responsibility. Data handling complies with UK GDPR and institutional ethics guidelines.

Professional, social, and economic implications

- Productivity improvements free accountants for higher-value work.
- Role shifts toward judgement and advisory tasks.
- Ethical deployment requires transparency and accountability.

Legal and compliance

- Adherence to IFRS/UK GAAP disclosure standards.
- GDPR-compliant anonymisation and consent.
- Final liability lies with the professional reviewer.

9. Resource Requirements

- Xero sandbox access or anonymised data.
- LLM API or local model environment.

- PostgreSQL, Python environment, and basic web hosting.
- Access to professional accountants for evaluation.

10. References / Information Sources

- Brynjolfsson, E., & McAfee, A. (2017). *The Business of Artificial Intelligence*. Harvard Business Review.
- Kokina, J., & Davenport, T. H. (2017). *The emergence of AI in accounting*. Journal of Emerging Technologies in Accounting.
- Issa, H., Sun, T., & Vasarhelyi, M. (2016). *Research Ideas for Artificial Intelligence in Auditing*.
- Mittelstadt, B. D. et al. (2016). *The ethics of algorithms: Mapping the debate*. Big Data & Society.
- Doshi-Velez, F., & Kim, B. (2017). *Towards a rigorous science of interpretable machine learning*. arXiv preprint.
- IFRS and UK GAAP disclosure standards.
- Xero developer documentation.

Appendix 1 — Project milestones (Gantt / activity plan)

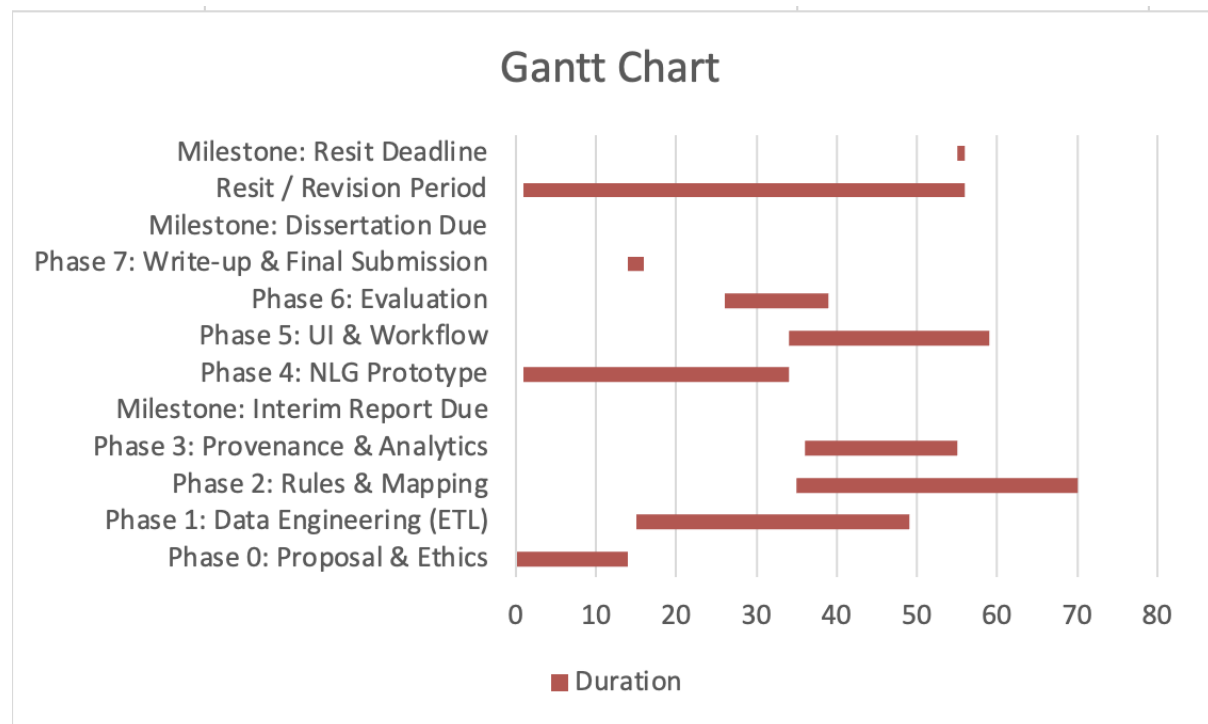


Figure 1: Gantt Chart